

Sergio Junior Ambrosio Camayo

Civil Engineer — professional degree, Universidad Nacional de Ingeniería (UNI), Faculty of Civil Engineering

Structural Engineer | Steel, Concrete & Mining | Design Codes & Advanced Analysis | Python & Engineering Automation

Portfolio · LinkedIn · GitHub

Profile

Civil engineer specializing in steel and concrete structures for buildings, mining and electrical infrastructure. I combine design-code criteria, modelling and advanced analysis with structural design, verification and technical documentation. Python and engineering automation support results processing, parametric studies and consistency checks. I currently work on structural engineering for mining projects at SRK, with a focus on justified decisions and traceable deliverables.

Professional experience

SRK Structural engineering for mining projects • Design of steel and concrete structures. • Structural modelling, analysis and verification. • Engineering work for mining infrastructure.	Present
HMV Ingenieros Structural Engineer — Professional III • Design of transmission towers, substation gantries and electrical equipment supports. • Modelling and verification of steel structures and their foundations. • Technical review of models, calculation reports, drawings and engineering deliverables. • For CEPITOCÉ, development of design criteria, assessment of design alternatives and technical coordination.	2026
MRZ Ingenieros Structural Engineer • Modelling and seismic analysis in ETABS of buildings with and without basements. • Design and detailing of beams, columns, structural walls, slabs and foundations; foundation analysis in SAFE. • Design of steel structures, industrial buildings and roof trusses. • Assessment of jacketing and FRP strengthening solutions. • Technical reviews, structural design optimization, calculation reports and drawings. • Coordination of structural designs with architecture and MEP systems.	November 2018 — January 2026
AMEC Foster Wheeler Perú Engineering Assistant • Structural inspections. • Topographic surveying. • Preparation of inventories and quantity take-offs.	March — July 2018
Grupo MIMCO S.A.C. Basic Engineering Assistant • Contribution to the design of telecommunications towers and monopoles. • Preparation of calculation reports.	January 2017 — March 2018

Selected projects

SUNSETGOLF TORRE C · 2024

Structural modelling and design of a 21-storey building with 3 basements: verification of primary members and foundations, with architecture and MEP coordination.

CEPITOCE / ELECTRICAL INFRASTRUCTURE · 2026

Design and verification of towers, gantries and equipment supports for high-voltage infrastructure, with assessment of design alternatives and engineering documentation.

ALMACENES CIDELSA / YANACOCCHA · 2024

Design and verification of an approximately 20 × 30 m steel structure, including primary members, connections and foundations, with architecture and MEP coordination.

Research and publications

PERFORMANCE OF SEISMICALLY ISOLATED LIQUID STORAGE TANKS USING TRIPLE PENDULUM BEARINGS

S. J. Ambrosio; E. J. Mamani; V. I. Fernández-Dávila · 2021

17th World Conference on Earthquake Engineering · Sendai, Japan · 2021

PERFORMANCE OF SEISMICALLY ISOLATED WATER STORAGE TANKS

E. J. Mamani; S. Ambrosio; V. I. Fernández-Dávila · 2018

XXXVIII South American Conference on Structural Engineering · Lima · 2018

DESEMPEÑO SÍSMICO DE TANQUES PARA ALMACENAMIENTO DE AGUA CON BASE AISLADA

Sergio J. Ambrosio; Elvis J. Mamani; Víctor I. Fernández-Dávila · 2018

12th International Symposium on Disaster Risk Management · Lima · 2018

Expertise and tools

Reinforced concrete · Steel structures · Earthquake engineering · Nonlinear analysis · Finite element analysis · Computational engineering · Assessment and strengthening · Modelling and documentation

ETABS · SAFE · Mathcad · AutoCAD · SAP2000 · IDEA StatiCa · Revit · Tekla Structures · Python · OpenSeesPy · Abaqus · AutoLISP

Codes and verification: RNE E.020 / E.030 / E.050 / E.060; ACI 318, ASCE 7, AISC 360, ASCE 41, ACI 562. Scope and reference editions.

Python / AutoLISP: results processing, validation and documentation. New portfolio demonstrations (2026): CSV / SI · parametric cantilever.

Education

Civil Engineer — professional degree, Universidad Nacional de Ingeniería, Faculty of Civil Engineering.

Civil engineering studies: 2013–2017. Upper third of the class, as reported in the CV.